

AE 301 Final Project

AE: 301 Low Speed Aerodynamics.

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Introduction

For our final project in AE 301 Low Speed Aerodynamics at San Diego State, Professor Katz wants us to show that we really understand how low speed airfoils work and how they are designed. In this report I am focusing on the GA(W)-1 airfoil and looking at some basic aerodynamic ideas like lift, drag, angle of attack, how the lift is spread along the wing, and what kind of bending loads show up at the root.

In the Results and Discussion section, I will use MATLAB to make the required plots for speeds from 35 to 130 meters per second. These graphs will help show how the airfoil behaves, how the airplane performs at different speeds, and what that means for things like stall speed, efficiency, and overall wing loading. Names of the graphs that will be shown are:

Figure 1: Airfoil Lift

Figure 2: Airfoil Drag

Figure 3: Angle of attack versus airspeed

Figure 4: Lift coefficient versus airspeed

Figure 5: Drag coefficient versus airspeed

Figure 6: Lift to drag ratio versus airspeed

Figure 7: Power required for the aircraft to maintain flight versus airspeed

Nomenclature:

$L = \text{lift} [N]$

$D = \text{drag} [N]$

$W = \text{weight [N]}$

$C_L = \text{coefficient of lift}$

$C_{L\alpha} = \text{lift curve slope}$

$C_D = \text{coefficient of drag}$

$C_{Di} = \text{coefficient of induced drag}$

$AR = \text{aspect ratio}$

$S = \text{wing area [m}^2\text{]}$

$b = \text{wing span [m]}$

$\rho = \text{density [kg/m}^3\text{]}$

$U = \text{speed [m/s]}$

$\alpha = \text{angle of attack}$

$\alpha_{L0} = \text{Zero Lift Angle of Attack}$

$P = \text{power [HP]}$

$M = \text{bending moment [N} \cdot \text{m]}$

$Re = \text{Reynolds number}$

$e = \text{span efficiency factor}$

$q = \text{dynamic pressure (N/m}^2\text{)}$

$RBM = \text{root bending moment}$

Equations:

$$AR = \frac{b^2}{S} \quad \text{equation 1}$$

$$L = W = \frac{1}{2} \rho U^2 S C_L \Rightarrow C_L = \frac{2W}{\rho U^2 S} \quad \text{equation 2}$$

$$D = \frac{1}{2} \rho U^2 S C_D \Rightarrow C_D = \frac{2D}{\rho U^2 S} \quad \text{equation 3}$$

$$C_{L\alpha} = \frac{2\pi}{1 + \frac{2}{AR}} \quad \text{if slope is } 2\pi \quad \text{equation 4}$$

$$C_{L\alpha} = \frac{\alpha_0}{1 + \frac{\alpha_0}{\pi A Re}} \quad \text{equation 5}$$

$$C_{Di} = \frac{S}{\pi b^2} C_L^2 \quad \text{equation 6}$$

$$C_D = C_{D,2D} + \Delta C_D + C_{Di} \quad \text{equation 7}$$

$$P = DU \quad \text{equation 8}$$

$$q = \frac{1}{2} \rho U^2 \quad \text{equation 9}$$

$$D = q S C_D \quad \text{equation 10}$$

$$RBM = \frac{Lb}{3\pi} \quad \text{equation 11}$$

$$RBM = -\frac{\pi}{4} \rho U^2 \int_0^c x \frac{d}{dx} \left[\alpha \frac{x^2}{c^2} b^2 \right] dx = -\frac{\pi}{4} \rho U^2 \alpha b^2 \frac{2c}{3} = -L \frac{2c}{3}$$

equation 12

Results and discussion

Given quantities-

Design: GA(W) – 1 Airfoil

Mass: 2 metric tons

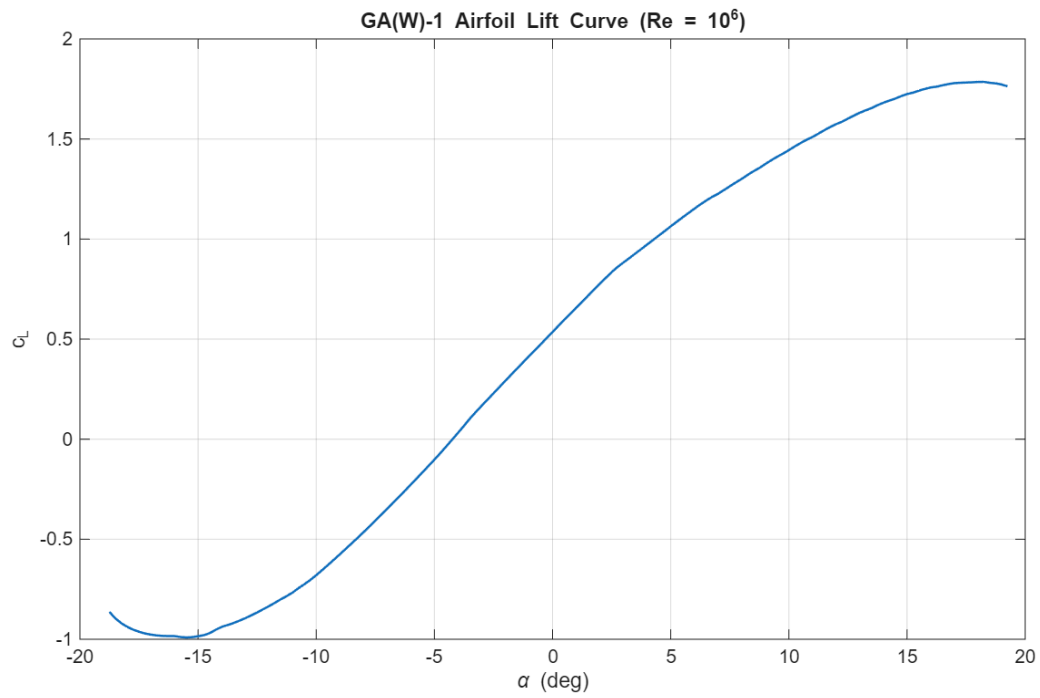
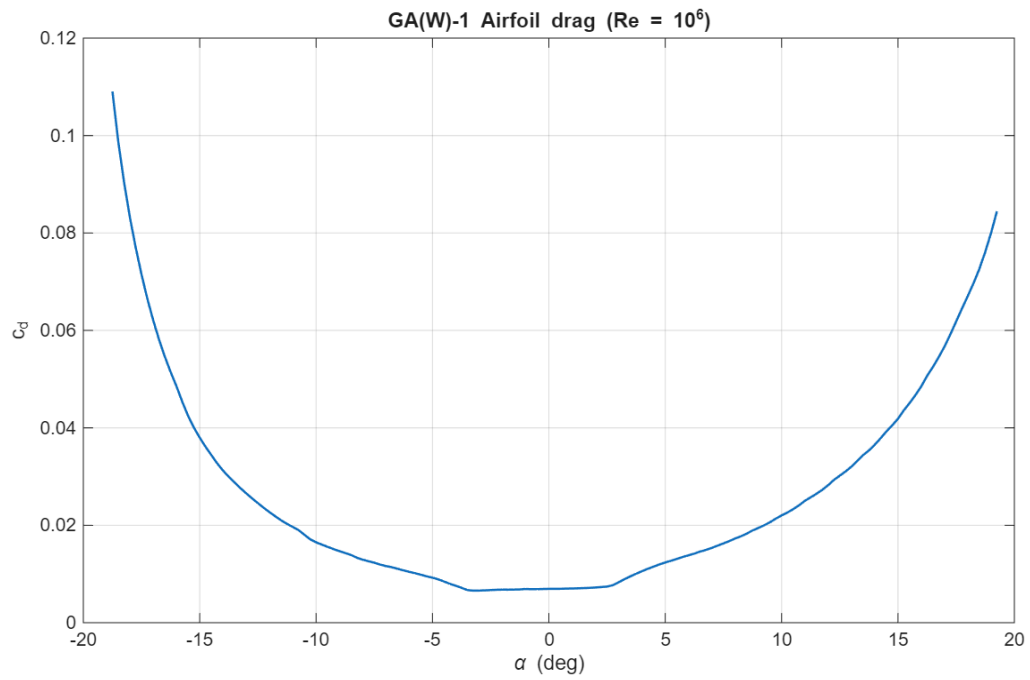
Wing span: $b = 13.2 \text{ m}$

Wing area: $S = 16.25 \text{ m}^2$

Additional drag: $\Delta C_D = 0.012$

Density: $\rho = 1.2 \text{ kg/m}^3$

*Reynolds number: $Re = 1 * 10^6$*

*Figure 1: airfoil lift**Figure 2: airfoil drag*

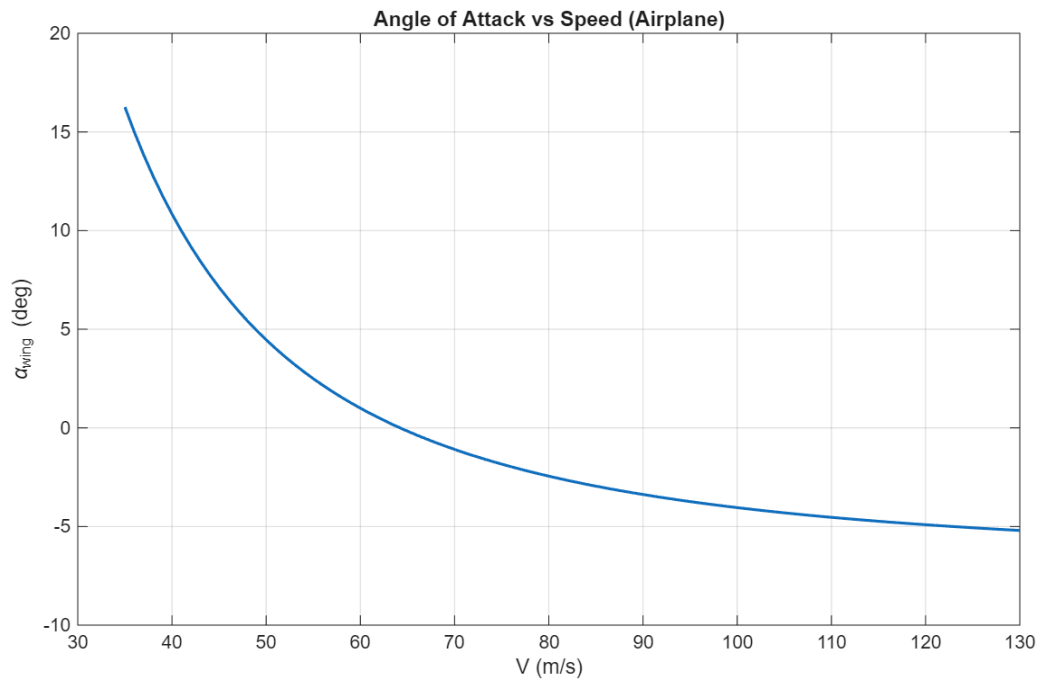


Figure 3: angle of attack vs speed

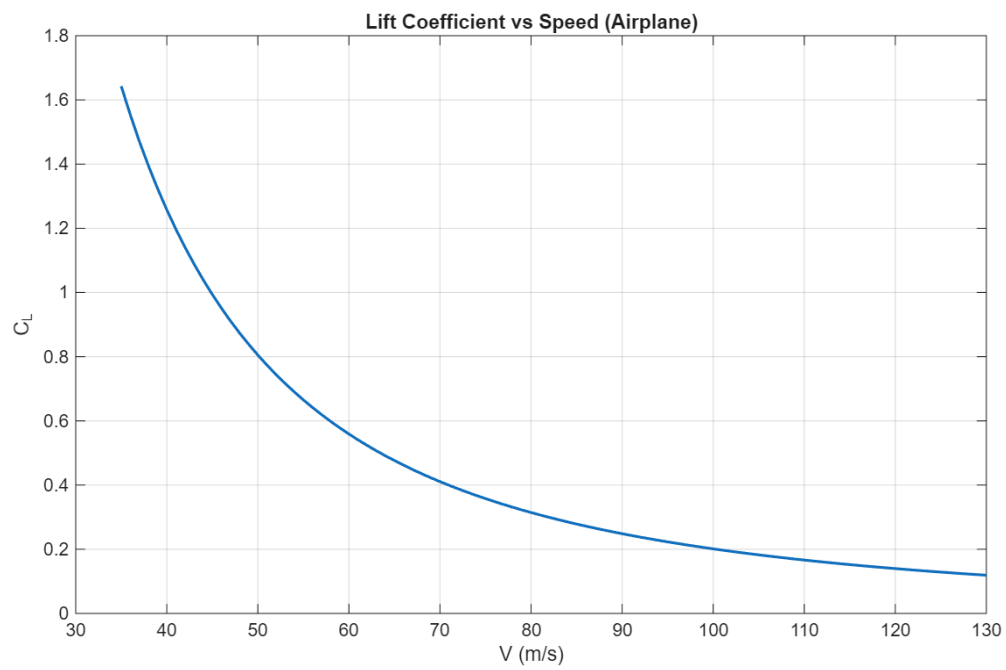


Figure 4: lift coefficient vs speed

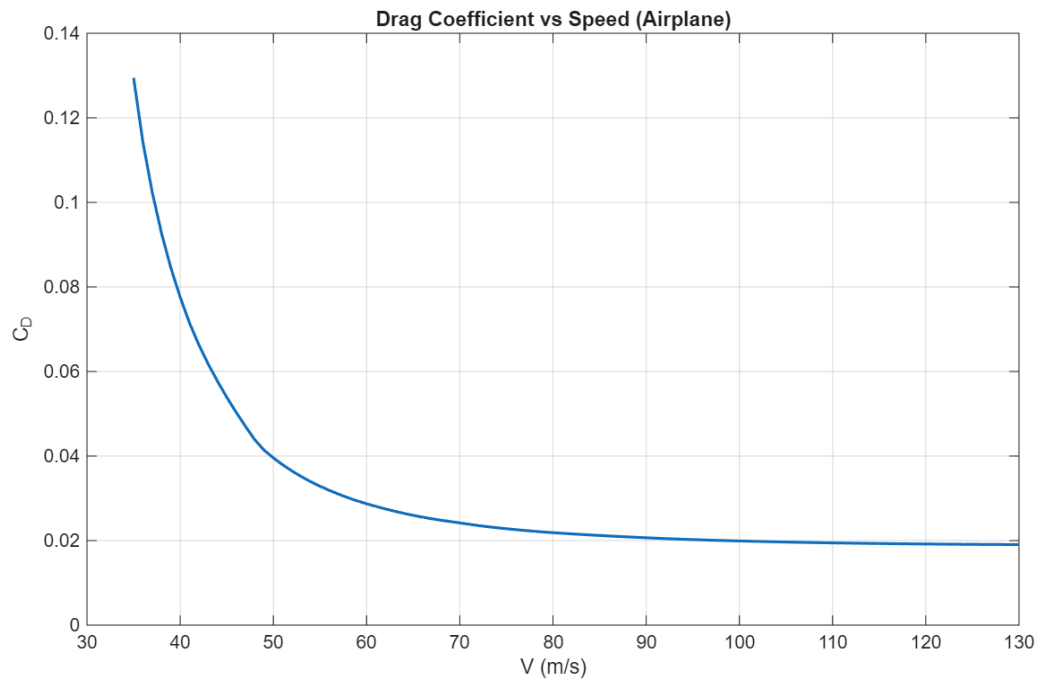


Figure 5: drag coefficient vs speed

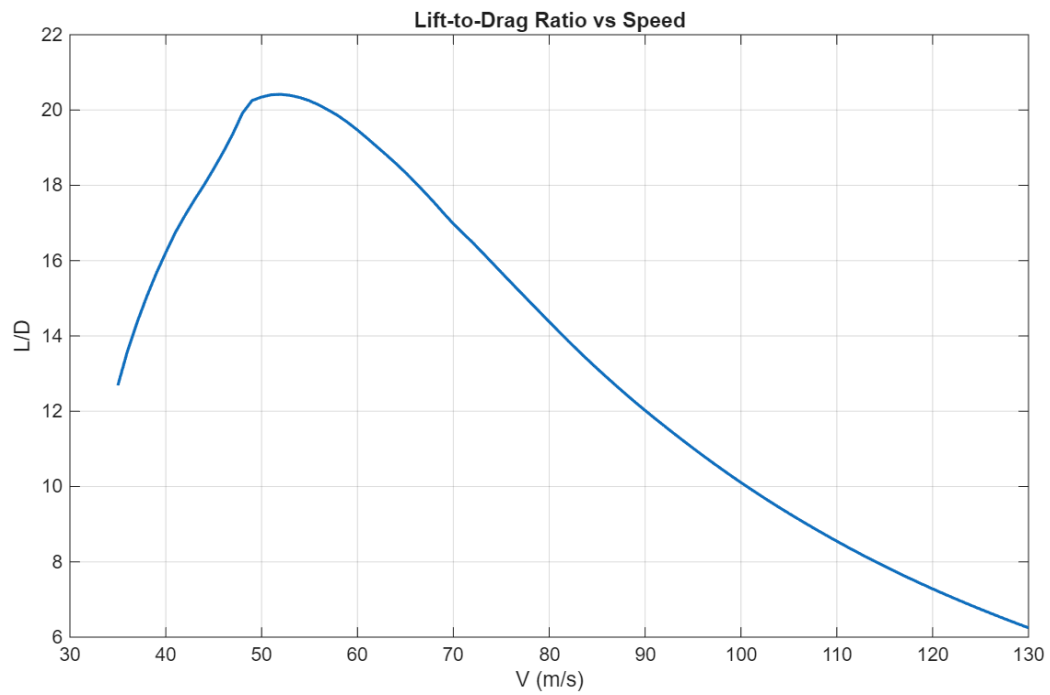


Figure 6: lift/drag ratio vs speed

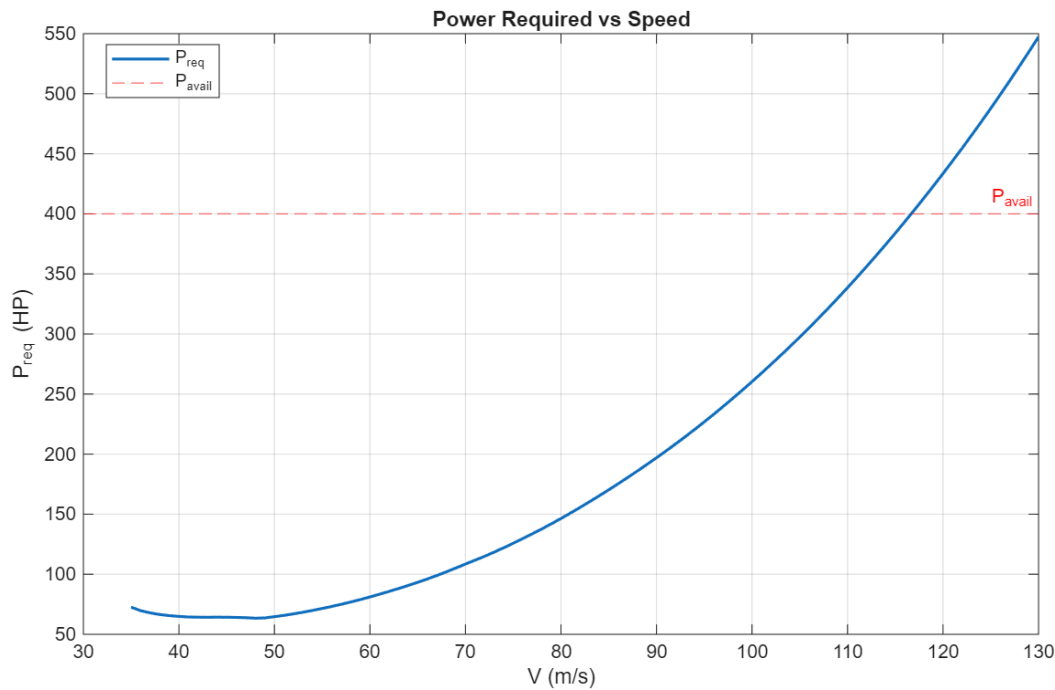


Figure 7: power required to propel the airplane vs speed

All seven required plots were successfully generated using the MATLAB code provided in the annex. The first two plots, C_l versus angle of attack and C_d versus angle of attack, are fundamental tools for evaluating airfoil performance. The lift coefficient plot displays the expected nearly linear increase followed by a sharp drop, which corresponds to the stall angle or critical angle of attack. The drag coefficient plot also matches the typical behavior seen in aerodynamic data, and in this case the maximum C_d is approximately 0.11.

The remaining five plots illustrate angle of attack, lift coefficient, drag coefficient, lift to drag ratio, and power as functions of speed. These graphs are essential, since they form the basis for the performance calculations discussed later. The C_l and C_d plots versus speed exhibit similar trends because of their mathematical dependence on velocity; both vary inversely with the square of airspeed, with the difference being the relative magnitude of lift and drag. The lift to drag ratio

plot was used to determine the most efficient operating speed, since the peak of this curve indicates the maximum aerodynamic efficiency. Finally, the power versus speed plot was used to identify the maximum possible speed. Given the specified available power, the curve approaches an asymptote where the required power equals the available power, which allows the maximum achievable speed to be identified from the graph.

Calculations for stall speed, most efficient speed, max speed, and RBM :

Stall Speed:

The stall speed is the minimum airspeed an aircraft must exceed in order to maintain controlled flight. It is not a fixed value, since it depends on the angle of attack. When the angle of attack becomes too large for a given airspeed, the airflow can no longer remain attached to the upper surface of the wing. The flow separates, becomes highly turbulent, and the wing experiences a sharp loss of lift along with a significant rise in drag, which leads to an uncommanded descent. As indicated by equation 2, airspeed and lift coefficient vary inversely, so an increase in one corresponds to a decrease in the other. For this reason, the maximum lift coefficient is used when determining the stall speed, because the aircraft reaches the stall condition when the lift coefficient can no longer increase to balance the required lift. Using the

maximum value provides the lowest possible speed at which the wing can still generate enough lift to sustain flight.

$$U_{stall} = \sqrt{\frac{2W}{\rho S C_{L,max}}} = \sqrt{\frac{(2)(2000)(9.8)}{(1.2)(16.25)(1.786)}}$$

$$U_{stall} = 33.57 \text{ m/s}$$

Most efficient speed:

The most efficient flight speed occurs at the maximum lift to drag ratio. In this condition, the aircraft generates the greatest amount of lift for the least amount of drag, which represents the point of peak aerodynamic efficiency. Flying at this speed minimizes the power or thrust required for steady, level flight and is therefore important when evaluating performance metrics such as thrust to weight ratio or endurance. From the Lift to Drag ratio versus airspeed plot, the maximum $\frac{L}{D}$ value obtained from MATLAB was 20.42. This peak corresponds to an optimal operating speed of approximately 52 m/s.

$$U_{efficient} = 52 \text{ m/s}$$

Max speed, max power (400 hp):

The maximum speed is the highest airspeed the aircraft can achieve given the available engine power. In this analysis, it is determined using equations 8, 9, and 10, since each equation provides quantities needed to evaluate the next. Because drag acts in the direction opposite to the motion, it is the aerodynamic force directly associated with the power required for flight. In this case, both the power available and the drag force are either provided or computed from the

relevant parameters, which allows the maximum achievable speed to be determined.

$$q = 1/2\rho U^2$$

$$D = qSC_D$$

$$P_{max} = DU_{max} \Rightarrow U_{max} = \frac{P_{max}}{D}$$

$$U_{max} = 116.5 \text{ m/s}$$

Root Bending Moment:

The root bending moment in an airfoil is the bending and twisting load that develops at the wing root, where the wing attaches to the fuselage. This moment results from the lift produced by the wing and the way that lift is distributed along the span. From the class slides, equation 11 was derived to express the root bending moment in terms of the total lift and the wing span. The formulation applies to a single wing panel rather than both sides of the aircraft. In this analysis, the load is taken to be the weight supported by the wing, which allows it to be used directly in place of the lift when computing the root bending moment.

$$RBM = \frac{Lb}{3\pi} = \frac{Wb}{3\pi} = \frac{(2000)(9.8)(13.2)}{3\pi}$$

$$RBM = 27479.1 \text{ Nm}$$

Conclusion:

In conclusion, the GA(W)-1 airfoil was successfully evaluated and graphically analyzed throughout this report. The GA(W)-1, developed by NASA for general aviation applications, is designed to provide a balance of low drag during cruise and high lift capability. Based on the analysis across the speed range of 35 to 130 meters per second, the results support this intended performance. An assumption of elliptical loading was applied throughout the calculations, which helped simplify the analysis while still providing realistic estimates of aerodynamic behavior. Overall, the development of the MATLAB code, the generation of the required plots, and the interpretation of the resulting data were made possible through the concepts and methods taught in AE 301, Low Speed Aerodynamics at San Diego State University.

Appendices:

Matlab Codes:

Code for figures 1-7 ↓

```
%% 1. Load GA(W)-1 polar (Re = 1e6)
dataTable = readtable('gaw1_polar.csv');
alpha_deg = dataTable.Alpha;
cl_section = dataTable.Cl;
cd_section = dataTable.Cd;
%% 2. 2D lift curve fit
```

```

alpha_range = (alpha_deg >= -4) & (alpha_deg <= 18);
fit_coeff    = polyfit(alpha_deg(alpha_range), cl_section(alpha_range), 1);
a0_deg       = fit_coeff(1);                    % 2D lift slope per deg
alphaL0_deg  = -fit_coeff(2) / fit_coeff(1);    % zero lift AoA (deg)
a0_rad       = a0_deg * 180/pi;

%% 3. Geometry and constants
mass         = 2000;                          % kg
g            = 9.81;                          % m/s^2
W            = mass * g;                      % N
b            = 13.2;                          % span [m]
S            = 16.25;                         % planform area [m^2]
rho          = 1.2;                          % kg/m^3
AR           = b^2 / S;                      % aspect ratio
e            = 0.95;                          % Oswald efficiency
deltaCD0     = 0.012;                        % extra parasite drag (airframe)
hp_to_W      = 745.7;

P_avail_hp   = 400;                          % available power in HP
P_avail      = P_avail_hp * hp_to_W;         % available power in W

% Finite wing lift curve slope (same formula, just different variable names)
CLalpha_rad = a0_rad / (1 + a0_rad/(pi * e * AR));
CLalpha_deg = CLalpha_rad * pi/180;

%% 4. Build Cd(CL) from 2D polar
[cl_sort, idx_sort] = sort(cl_section);
cd_sort             = cd_section(idx_sort);
[cl_unique, idx_uniq] = unique(cl_sort);
cd_unique           = cd_sort(idx_uniq);
CL_max = max(cl_unique);

%% 5. Level flight CL vs speed (35-130 m/s)
V = (35:1:130)';                    % speed vector [m/s]
q = 0.5 * rho .* V.^2;              % dynamic pressure
CL_req = W ./ (q * S);              % lift coefficient required for level flight
% Stall speed using section CL_max

```

```

V_stall = sqrt(2*W / (rho * S * CL_max));
% Replace CL below stall with NaN
below_stall      = V < V_stall;
CL_req(below_stall) = NaN;
%% 6. AoA(V), Cd(V), L/D(V), power required
% AoA using finite-wing CL-alpha
alpha_wing_deg = alphaL0_deg + CL_req ./ CLalpha_deg;
% Profile drag from polar (as function of CL)
Cd_profile = interp1(cl_unique, cd_unique, CL_req, 'pchip', NaN);
% Induced drag
Cd_ind = CL_req.^2 ./ (pi * e * AR);
% Total airplane drag coefficient
CD_total = Cd_profile + deltaCD0 + Cd_ind;
% L/D and drag and power
LD      = CL_req ./ CD_total;
D      = q .* S .* CD_total;      % drag [N]
P_req   = D .* V;                  % required power [W]
P_req_hp = P_req / hp_to_W;        % required power [HP]
%% 7. Performance metrics
% Max L/D and corresponding speed
[LD_max, idx_LDmax] = max(LD);
V_LDmax = V(idx_LDmax);
% Extend speed range for max speed with available power
V_ext = (35:0.5:200)';
q_ext = 0.5 * rho .* V_ext.^2;
CL_ext = W ./ (q_ext * S);
CL_ext(V_ext < V_stall) = NaN;
alpha_ext_deg = alphaL0_deg + CL_ext ./ CLalpha_deg;
Cd_prof_ext   = interp1(cl_unique, cd_unique, CL_ext, 'pchip', NaN);
Cd_ind_ext    = CL_ext.^2 ./ (pi * e * AR);
CD_ext        = Cd_prof_ext + deltaCD0 + Cd_ind_ext;
D_ext         = q_ext .* S .* CD_ext;

```

```

P_req_ext      = D_ext .* V_ext;

% Max speed where power required <= power available
feasible_idx = find(P_req_ext <= P_avail & ~isnan(P_req_ext));
if ~isempty(feasible_idx)
    V_max = V_ext(max(feasible_idx));
else
    V_max = NaN;
end

% Simple root bending moment (1 g)
M_root = W * b / (3*pi);

%% 8. Text summary
fprintf('\n=== AE301 Wing Performance Summary ===\n');
fprintf('Zero-lift alpha (deg)           : %7.3f\n', alphaL0_deg);
fprintf('2D lift slope a_0 (per deg)       : %7.4f\n', a0_deg);
fprintf('3D lift slope CL_alpha (deg^-1): %7.4f\n', CLalpha_deg);
fprintf('Section CL_max                     : %7.3f\n', CL_max);
fprintf('Stall speed V_stall                 : %7.2f m/s\n', V_stall);
fprintf('Speed at max L/D                    : %7.2f m/s, L/D = %7.2f\n', V_LDmax, LD_max);
fprintf('Max speed with %g HP                : %7.2f m/s\n', P_avail_hp, V_max);
fprintf('Root bending moment (1 g)          : %7.1f N·m\n\n', M_root);

%% 9. Plots for report
% Airfoil lift curve (c_l vs alpha)
figure;
plot(alpha_deg, cl_section, 'LineWidth', 1.2);
grid on;
xlabel('\alpha (deg)');
ylabel('c_L');
title('GA(W)-1 lift data (Re = 10^6)');

% Airfoil drag data (c_d vs alpha)
figure;
plot(alpha_deg, cd_section, 'LineWidth', 1.2);

```



```
grid on;
xlabel('\alpha (deg)');
ylabel('c_d');
title('GA(W)-1 drag data (Re = 10^6)');
% Angle of attack vs speed
figure;
plot(V, alpha_wing_deg, 'LineWidth', 1.5);
grid on;
xlabel('V (m/s)');
ylabel('\alpha_{wing} (deg)');
% title intentionally left out so graph looks slightly different
% CL vs speed
figure;
plot(V, CL_req, 'LineWidth', 1.5);
grid on;
xlabel('V (m/s)');
ylabel('C_L');
title('Required C_L vs airspeed');
% CD vs speed
figure;
plot(V, CD_total, 'LineWidth', 1.5);
grid on;
xlabel('V (m/s)');
ylabel('C_D');
title('Total C_D vs airspeed');
% L/D vs speed
figure;
plot(V, LD, 'LineWidth', 1.5);
grid on;
xlabel('V (m/s)');
ylabel('L/D');
title('L/D variation with speed');
```

```
% Power required vs speed
```

```
figure;
```

```
plot(V, P_req_hp, 'LineWidth', 1.5);
```

```
hold on;
```

```
ylines(P_avail_hp, 'r--', 'P_{avail}');
```

```
grid on;
```

```
xlabel('V (m/s)');
```

```
ylabel('P_{req} (HP)');
```

```
title('Power requirement curve');
```

```
legend('P_{req}', 'P_{avail}', 'Location', 'best');
```

```
hold off;
```

Alpha	Cl	Cd	Cdp	Cm	Top_Xtr	Bot_Xtr		
-18.75	-0.8636		0.10904		0.10626	-0.0535	1	0.0155
-18.5	-0.8958		0.09877		0.0958	-0.0596	1	0.0154
-18.25	-0.9195		0.0904	0.08726		-0.0647	1	0.0154
-18	-0.938	0.08321		0.07992		-0.0691		0.0155
-17.75	-0.9517		0.077	0.07358		-0.0728	1	0.0155
-17.5	-0.9625		0.07147		0.06791	-0.0761		0.0155
-17.25	-0.9709		0.0665	0.06281		-0.0789	1	0.0156
-17	-0.9767		0.06211		0.0583	-0.0814	1	0.0156
-16.75	-0.9808		0.05816		0.05423	-0.0834	1	0.0157
-16.5	-0.9832		0.05463		0.05059	-0.0851	1	0.0157
-16.25	-0.9842		0.05145		0.0473	-0.0865	1	0.0158
-16	-0.9842		0.04856		0.04432	-0.0875	1	0.0158
-15.75	-0.9887		0.04535		0.04097	-0.0881		0.0159
-15.5	-0.991	0.04246		0.03796		-0.0881		0.016
-15.25	-0.9895		0.04005		0.03547	-0.0881		0.0161
-15	-0.985	0.03798		0.03333		-0.0881		0.0162
-14.75	-0.9792		0.03611		0.03142	-0.0879	1	0.0163
-14.5	-0.969	0.03441		0.02966		-0.0883	0.9951	0.0164
-14.25	-0.9523		0.0327	0.02789		-0.0898	0.9818	0.0165
-14	-0.9383		0.03123		0.02635	-0.0903	0.9655	0.0167

-13.75-0.9288	0.02994	0.02498	-0.0896	0.948	0.0168
-13.5 -0.9189	0.02876	0.0237-0.0887	0.9337	0.0169	
-13.25-0.9072	0.02762	0.02247	-0.0880.9219	0.017	
-13 -0.8944	0.02654	0.0213-0.0874	0.9122	0.0172	
-12.75-0.8806	0.02552	0.02019	-0.0869	0.9031	0.0173
-12.5 -0.8660.02453	0.01912	-0.0864	0.8955	0.0175	
-12.25-0.8503	0.02358	0.0181-0.0860.8885	0.0177		
-12 -0.8346	0.02268	0.01711	-0.0855	0.8814	0.0178
-11.75-0.8178	0.02183	0.01619	-0.0850.8752	0.018	
-11.5 -0.8005	0.02101	0.01532	-0.0846	0.8692	0.0182
-11.25-0.7843	0.02028	0.01451	-0.0839	0.8632	0.0184
-11 -0.7665	0.01964	0.01379	-0.0834	0.8574	0.0186
-10.75-0.7445	0.019	0.01311	-0.0836	0.8535	0.0187
-10.5 -0.7247	0.01804	0.0121-0.0838	0.849	0.019	
-10.25-0.7037	0.01711	0.01112	-0.0841	0.8444	0.0194
-10 -0.6798	0.01650.01046	-0.0845	0.84	0.0197	
-9.75 -0.6545	0.01602	0.00995	-0.0849	0.8362	0.0201
-9.5 -0.6284	0.01556	0.00947	-0.0853	0.8332	0.0206
-9.25 -0.6020.01511	0.009	-0.0857	0.8298	0.021	
-9 -0.5752	0.01469	0.00854	-0.0861	0.8264	0.0214
-8.75 -0.5480.01431	0.00811	-0.0865	0.823	0.0218	
-8.5 -0.5207	0.01391	0.00766	-0.0870.8192	0.0222	
-8.25 -0.4935	0.01334	0.00707	-0.0877	0.816	0.0229
-8 -0.4651	0.01295	0.00668	-0.0883	0.8134	0.0236
-7.75 -0.4362	0.01261	0.00633	-0.0889	0.8106	0.0244
-7.5 -0.4071	0.01231	0.00601	-0.0896	0.8078	0.0252
-7.25 -0.3779	0.01194	0.00562	-0.0903	0.8051	0.0263
-7 -0.3484	0.01162	0.00529	-0.0910.8023	0.0277	
-6.75 -0.3185	0.01139	0.00502	-0.0917	0.7991	0.0293
-6.5 -0.2884	0.01107	0.00473	-0.0926	0.7966	0.0325
-6.25 -0.2580.01074	0.00447	-0.0935	0.7946	0.0401	
-6 -0.2273	0.01044	0.00425	-0.0944	0.7924	0.0539

-5.75	-0.1964	0.01016	0.00402	-0.0954	0.79010.065	
-5.5	-0.1653	0.00985	0.00381	-0.0964	0.78770.0814	
-5.25	-0.1339	0.00952	0.0036	-0.0976	0.78520.1097	
-5	-0.1023	0.00924	0.00342	-0.0987	0.78270.1338	
-4.75	-0.07	0.00889	0.00323	-0.1002	0.78	0.1749
-4.5	-0.0366	0.00845	0.00304	-0.1020	0.77740.2379	
-4.25	-0.0025	0.00797	0.00285	-0.1040	0.77590.3092	
-4	0.0316	0.00759	0.00269	-0.1058	0.77410.3698	
-3.75	0.0665	0.00716	0.00252	-0.1078	0.77190.4425	
-3.5	0.1025	0.00669	0.0024	-0.1101	0.76970.5372	
-3.25	0.1353	0.00658	0.00243	-0.1112	0.76740.59	
-3	0.1668	0.00659	0.00246	-0.1120	0.76510.6097	
-2.75	0.198	0.00664	0.00248	-0.1126	0.76280.6218	
-2.5	0.2291	0.00669	0.00251	-0.1133	0.76	0.6294
-2.25	0.2601	0.00676	0.00257	-0.1139	0.75740.6356	
-2	0.2914	0.00679	0.00258	-0.1146	0.75570.642	
-1.75	0.3222	0.00678	0.0026	-0.1152	0.753	0.6485
-1.5	0.3529	0.00680	0.00262	-0.1157	0.74930.6537	
-1.25	0.3836	0.00683	0.00262	-0.1162	0.74470.6597	
-1	0.4139	0.00692	0.00265	-0.1166	0.73840.6661	
-0.75	0.4441	0.00688	0.00266	-0.1169	0.73250.6711	
-0.5	0.4745	0.00688	0.00264	-0.1174	0.72630.674	
-0.25	0.5049	0.00692	0.00264	-0.1179	0.72150.6766	
0	0.5356	0.00695	0.00266	-0.1184	0.71710.6789	
0.25	0.5664	0.00695	0.00267	-0.1190	0.71230.6808	
0.5	0.5969	0.00696	0.00267	-0.1196	0.70760.6839	
0.75	0.6267	0.007	0.00269	-0.1199	0.70140.6881	
1	0.6568	0.00702	0.00274	-0.1203	0.69410.6914	
1.25	0.6866	0.00706	0.00276	-0.1206	0.68520.694	
1.5	0.7167	0.00710	0.0028	-0.1211	0.67710.6964	
1.75	0.7467	0.00715	0.00283	-0.1215	0.66680.6985	
2	0.7766	0.00722	0.00286	-0.1219	0.65530.7002	

2.25	0.80580.00732	0.0029	-0.1222	0.63920.7014
2.5	0.83450.00741	0.00293	-0.1224	0.61660.7035
2.75	0.86030.00770.00304	-0.1221	0.57180.7055	
3	0.88280.00829	0.00332	-0.1213	0.50570.7073
3.25	0.90470.00894	0.00367	-0.1204	0.444 0.709
3.5	0.92740.00950.00399	-0.1196	0.39320.7105	
3.75	0.94960.01006	0.00431	-0.1188	0.346 0.7122
4	0.97230.01057	0.00463	-0.1180.30620.7138	
4.25	0.99490.01106	0.00494	-0.1173	0.27160.7155
4.5	1.01770.01151	0.00523	-0.1165	0.24170.7171
4.75	1.04020.01196	0.00555	-0.1157	0.21730.7188
5	1.06340.01235	0.00583	-0.1150.19760.7202	
5.25	1.08590.01272	0.00612	-0.1143	0.179 0.7223
5.5	1.108 0.01310.00643	-0.1134	0.16380.7242	
5.75	1.12980.01348	0.00675	-0.1124	0.152 0.726
6	1.15140.01384	0.00707	-0.1114	0.14260.7279
6.25	1.17280.01418	0.00739	-0.1104	0.13430.7298
6.5	1.19230.01460.00776	-0.1090.12560.7318		
6.75	1.21030.01494	0.00808	-0.1073	0.11930.7339
7	1.22590.01536	0.00848	-0.1052	0.11290.7359
7.25	1.24460.01581	0.0089-0.1038	0.10690.7377	
7.5	1.26350.01628	0.00935	-0.1025	0.10130.7397
7.75	1.28250.01674	0.00981	-0.1012	0.09690.7422
8	1.30040.01728	0.01035	-0.0999	0.09270.7445
8.25	1.32030.01773	0.01082	-0.0988	0.09010.7469
8.5	1.33860.01828	0.01137	-0.0976	0.08730.7495
8.75	1.35530.01893	0.01202	-0.0961	0.08420.7521
9	1.37460.01944	0.01256	-0.0951	0.08250.7546
9.25	1.39350.01999	0.01313	-0.0941	0.08050.7569
9.5	1.41110.02062	0.01378	-0.0929	0.07850.76
9.75	1.427 0.02136	0.01454	-0.0915	0.07630.7628
10	1.44440.02202	0.01524	-0.0904	0.07470.7656

10.25	1.46280.02262	0.01589	-0.0894	0.07330.7685
10.5	1.48010.02331	0.0166-0.0883	0.07180.7713	
10.75	1.49610.02410.0174-0.0872	0.07010.7741		
11	1.51 0.02502	0.01834	-0.0857	0.06820.7774
11.25	1.527 0.02574	0.01912	-0.0847	0.067 0.7806
11.5	1.54390.02647	0.01991	-0.0837	0.06580.784
11.75	1.55990.02728	0.02075	-0.0827	0.06430.7875
12	1.57420.02823	0.0217-0.0815	0.06290.7907	
12.25	1.58630.02932	0.02282	-0.0801	0.06120.7944
12.5	1.60160.03018	0.02375	-0.0790.06010.798	
12.75	1.61690.03106	0.02469	-0.0780.05890.8017	
13	1.63120.03203	0.02569	-0.0770.05730.8053	
13.25	1.64310.03319	0.02686	-0.0758	0.05580.8087
13.5	1.65440.03439	0.02811	-0.0745	0.05430.8125
13.75	1.66890.03535	0.02914	-0.0736	0.053 0.8163
14	1.68110.03653	0.03037	-0.0725	0.05140.8202
14.25	1.69190.03784	0.03169	-0.0714	0.04990.8238
14.5	1.70180.03926	0.03316	-0.0702	0.04840.8278
14.75	1.714 0.04049	0.03446	-0.0693	0.047 0.8321
15	1.72440.04187	0.03589	-0.0683	0.04550.8366
15.25	1.73160.04359	0.03764	-0.0671	0.044 0.8409
15.5	1.74140.04509	0.03921	-0.0662	0.04280.8455
15.75	1.75 0.04672	0.04091	-0.0653	0.04150.8504
16	1.75740.04851	0.04274	-0.0644	0.04020.8554
16.25	1.76140.05066	0.04493	-0.0634	0.039 0.8601
16.5	1.76850.05249	0.04686	-0.0625	0.03810.8655
16.75	1.77440.05452	0.04896	-0.0618	0.03720.8713
17	1.77910.05669	0.0512-0.0610.03620.877		
17.25	1.781 0.05919	0.05377	-0.0603	0.03540.8836
17.75	1.78330.06450.05925	-0.0591	0.03390.8989	
18	1.78520.06707	0.06194	-0.0587	0.03340.9091
18.25	1.78550.06974	0.06473	-0.0581	0.03270.927

18.5	1.78070.07266	0.06777	-0.0572	0.03221
18.75	1.77790.07614	0.07132	-0.0573	0.03171
19	1.77240.08001	0.07526	-0.0576	0.03121
19.25	1.76320.08443	0.07977	-0.0581	0.03071